

(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	μw	radio
Frequency (ν)	↑						↓
Energy (E)	↑						↓
Wavelength (λ)	<0.01	0.01-10	10-400	400-700	700-1e6	1e6-1e9	>1e9

• **vis** (violet · blue · green · yellow · orange · red)

(2) $c = \lambda \nu$ (Speed of light = wavelength × frequency)

$\lambda \downarrow / \nu \rightarrow$	Hz	kHz	MHz	GHz
m	3.00e8 (m/s)	3.00e5 (m·kHz)	3.00e2 (m·MHz)	3.00e-1 (m·GHz)
nm	3.00e17 (nm·Hz)	3.00e14 (nm·kHz)	3.00e11 (nm·MHz)	3.00e8 (nm·GHz)

(3) $E = h\nu = h(c/\lambda)$ (Energy = Planck × frequency)

$E \downarrow / \lambda, \nu \rightarrow$	λ (m)	λ (nm)	ν (Hz)
J / photon	1.9864e-25 / λ (J·m)	1.9864e-16 / λ (J·nm)	6.626e-34 × ν (J·s)
J / mol	1.1962e-1 / λ (J·m/mol)	1.1962e8 / λ (J·nm/mol)	3.9902e-10 × ν (J·s/mol)
kJ / mol	1.1962e-4 / λ (kJ·m/mol)	1.1962e5 / λ (kJ·nm/mol)	3.9902e-13 × ν (kJ·s/mol)

• Grams → mol → ×6.022e23 for photon **count**

(4) Quantum Principles

Planck	Energy emitted only in fixed amounts (quanta)
Einstein	Light arrives in packets (photons) of fixed energy
Heisenberg	Position and velocity not both knowable
Bohr	e ⁻ occupy fixed energy levels; explains line spectra

• Orbital = **region where an e⁻ is most likely found**

• $E = h\nu = h(c/\lambda) = R(1/n_f^2 - 1/n_i^2)$

• $n_i = \sqrt{1/(1/n_f^2 - hc/(R\lambda))}$; $n_f = \sqrt{1/(1/n_i^2 + hc/(R\lambda))}$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) de Broglie

$$\lambda = h/(mu)$$

λ (m) given $h/(m \cdot u)$; $m \downarrow / u \rightarrow$	m/s	km/s
kg	6.626e-34	6.626e-37
g	6.626e-31	6.626e-34
amu	3.9892e-7	3.9892e-10

• Answer is in m — **×1e9 for nm** (1 nm = 1e-9 m)

(6) Electron Configuration

- Fill in Aufbau order (below)
- Respect **s2 p6 d10 f14**
- Apply the exceptions
 - Aufbau = **increasing energy** →: 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d

Trend	→ across period	↓ down group
Z_{eff}	increases	~constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more favorable	less favorable

- Cr = [Ar]4s¹3d⁵ · Cu = [Ar]4s¹3d¹⁰** (half/full d wins)
- Impossible if any subshell is over-filled (3s³, 2p⁸)
- Faster than writing Aufbau: read it off the table — s-block (gp 1-2) → ns, p-block (13-18) → np, d-block → (n-1)d; the period number is n
- Can be a transition metal; ground state only

(7) Reading an Electron Configuration

Which element?

- Noble-gas core → its own Z: [He]2 [Ne]10 [Ar]18 [Kr]36 [Xe]54
- Add every superscript to it = total e⁻ = Z (neutral atom)
- Look up Z on the periodic table
 - Cross-check: outermost subshell → period = n, block → group

Paramagnetic vs diamagnetic

- Draw boxes for the outer subshell — **s 1 · p 3 · d 5 · f 7**
- Fill singly, parallel spins, before pairing (Hund)
- Count unpaired e⁻

• Any unpaired → **paramagnetic** · all paired → **diamagnetic**

• Full subshells (s², p⁶, d¹⁰) are always paired

Excited vs ground state

- Ground** = strict Aufbau order, no gaps
- Excited** = an e⁻ sits higher than Aufbau requires, leaving a **gap below it**; same total e⁻ count

Valence electrons

- Count e⁻ in the **highest** n (outer s + p) = the group number, 1A→1 ... 8A→8
- Everything below the highest n is **core**

(8) Ion Configuration

- Start from the neutral atom
- Anion**: add e⁻ into the outer p until it reaches a noble-gas configuration
- Cation**: remove from the **highest** n **first**
 - Transition metals lose **4s before 3d**
 - Exam ions are main-group, not transition metals

(9) Quantum Numbers, Penetration & Shielding

sublevel = subshell	s	p	d	f
$l \leq n-1$	0	1	2	3
m_l	0	-1...+1	-2...+2	-3...+3
orbitals ($2l+1$)	1	3	5	7
max e ⁻ (2 per orbital)	2	6	10	14

Symbol	Allowed	Meaning
n	1, 2, 3, ...	Principal energy level, shell (row #)
m_s	±1/2	spin , 2 e ⁻ per orbital (Pauli)

- energy** set by n and l **only** — m_l , m_s are degenerate
- energy**: **s < p < d < f** (within shell, same n) · across shells use **Aufbau** (6)
- 3f and 2d don't exist** (row 1 above) — check the sublevel is real before ranking it
- Shell n holds n^2 orbitals, $2n^2$ e⁻
- Penetration** — orbital reaching into core region is less shielded, so **lower energy**.
- Pauli exclusion** — no two e⁻ carry the same four quantum numbers → 2 per orbital, opposite spins
- Shielding** — only **core** (inner) e⁻ shield effectively; valence e⁻ shield each other poorly

(10) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

- Core e⁻ = total e⁻ - valence e⁻ (= the **previous noble gas**)
- $Z_{eff} \approx Z - \text{core } e^-$
 - Same period → same core, so the **rightmost** element wins
 - Poor valence shielding is why Z_{eff} rises across a period

(11) Rank Atomic Size

- Down a group **bigger** (new shell) · across a period **smaller** (Z_{eff} pulls in)
- ↑ = **lower-left** · ↓ = **upper-right**
- Group position beats period position when both apply
- cation < neutral < anion

(12) Rank Ionization Energy

- Across a period ↑ · down a group ↓ (opposite of radius)
- ↑ = **upper-right** (He ↑ of all) · ↓ = **lower-left**
- $IE_2 > IE_1$ always
- Big jump** appears once the first **core** e⁻ is removed

(13) Rank Electron Affinity

- **Most eager for an e⁻ = halogens, upper right** — this course writes it both as "most negative *EA*" and "largest positive *EA*"; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- ***EA* (X) and *IE* of X are the same magnitude** — F + e⁻ releases exactly what F → F costs

(14) Families & Diagonal Rule

1A	alkali metals	most reactive metals; ↑ down
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; ↓ down
8A	noble gases	inert — full octet

- **Diagonal rule** — an element resembles its down-right neighbor: Li~Mg · Be~Al · B~Si

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K₂O, MgO, Fe₂O₃, Cr₂O₃) · nonmetal oxide = **acidic** (P₄O₁₀, SO₂, CO₂, Cl₂O₇) · Al₂O₃ and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3- · 6A 2- · 7A 1-; criss-cross and reduce

- Same-group swap keeps the pattern: BaS → **CaO**

- 2A + 3F₂ → 2AF₃ ⇒ A is 3+ (**Al**) · 3Li + Z → Li₃Z ⇒ Z is 3-, so Mg gives **Mg₃Z₂**

Bond-type ID — ionic **and** covalent together = a salt of a **polyatomic ion** (NH₄NO₃, K₂SO₄); all-nonmetal = covalent only

Family reactions

- 2K(s) + 2H₂O(l) → 2KOH(aq) + H₂(g)
- H₂(g) + Cl₂(g) → 2HCl(g)

Odds & ends

- Lanthanides = 4f row (**Ce**) · actinides = 5f (U)
- Diatomic: H₂ N₂ O₂ F₂ Cl₂ Br₂ I₂
- Metallic character ↑₂ down, ↓₂ across → **lower-left** wins
- Isolectronic = matches a noble gas e⁻ count (K⁺, Cl⁻, Ca²⁺ = Ar)

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first

2. If charges tie, smaller ions win

- ↑ charge → ↑ **U** (dominant) · ↑ ionic radius → ↓ U

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

- Read *EN* off the chart — *EN* ↑ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)

2. Classify

~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- **"Most polar"** → ΔEN **alone** (largest difference wins)
- **"Strongest / shortest bond"** → **bond order FIRST**: triple > double > single. ΔEN only breaks ties between equal orders — a C≡N beats a high- ΔEN Si=O
- Larger ΔEN → stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

1. *FC* on every atom

2. Check ΣFC = the overall charge

3. Pick the structure with *FC*'s **nearest 0**

4. Put any -1 on the **more electronegative** atom

- Equivalent resonance forms → real bonds are the **average** (e.g. 1½)

(19) Lewis Structures x4

1. Total valence e⁻ — +1 per negative charge, -1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e⁻ onto the central atom
6. Central below 8 → pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e ⁻ per atom (H wants 2) — H ₂ O, CH ₄ , NH ₃ , CO ₂ , CH ₂ Cl ₂ , CH ₃ CH ₂ Cl
(b) Incomplete octet	Al, B, Be only — content below 8, do not add double bonds to fix. BF ₃ /BCl ₃ /AlI ₃ leave 6 e ⁻ on the center; BeCl ₂ leaves 4
(c) Expanded octet	Central atom period 3 or below (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — AsCl ₅ , XeI ₂ , SO ₄ ²⁻
(d) Organic	-OH alcohol · C=O aldehyde/ketone · -COOH carboxylic acid · -NH₂ amine

- Every C makes 4 bonds; a -COOH carbon carries one **=O** and one **-O-H**; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero *FC*, add double bonds to zero it (e.g. two S=O in SO₄²⁻)

(20) Photoelectric Effect (*hν* - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

$$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$$

Variable	Value
<i>h</i>	6.626e-34 J·s
<i>m_e</i>	9.11e-31 kg
<i>hc</i>	1.9864e-25 J·m
2 <i>hc/m_e</i>	4.3609e5 m ³ /s ²

Have	Want	Use
ν_{thr} or λ_{thr}	Φ	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
Φ	Φ per mol	$\Phi \times N_A$ (/1e3 for kJ/mol)
Φ, λ	<i>u</i>	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2KE/m_e}$
λ, λ_{thr}	<i>u</i>	2nd eq above (skips Φ and <i>KE</i>)

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)

2. $\Delta x \geq h/(4\pi \cdot \Delta p)$

- Result is the **minimum** uncertainty
- Rearrange the same way to solve for Δu
- Keep everything in kg, m, s

(22) Born-Haber Cycle

$$\Delta H^\circ_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), *IE* (+), bond dissociation (+), *EA* (usually -), *U* (-)
2. **Halve** the bond dissociation if you need only one atom
3. Chain: elements → gaseous atoms → gaseous ions → solid
4. Set the chain equal to ΔH°_f
5. Solve for the single unknown
 - Use *IE₂* as well when the cation is 2+
 - U runs ions(g) → solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
2. Σ BE(broken), always positive
3. Σ BE(formed), subtracted
4. $\Delta H = \Sigma$ broken - Σ formed
 - Bonds unchanged on both sides cancel — skip them
 - Negative ΔH = exothermic
 - For "heat released by *g* of X," convert *g* → mol and scale